ScamShield™ Cyber Safety Platform

Complete Data Flow Diagram (DFD) Specification Document

Comprehensive Multi-Level DFD Documentation (Level 0, Level 1, Level 2.1 - 2.4)
This specification details the structural data flow, external entity interfaces, process decompositions, and data store schemas for the ScamShield digital scam check system.

Document Version:	v2.4 (Production Grade)
Target Platform:	ScamShield Web, PWA & Android Ecosystem
Notation Standard:	Gane & Sarson DFD Methodology
Engine Components:	Python Flask, Cloudflare D1 SQL, Multi-Tier OCR, Risk Engine
Security Controls:	In-Memory Processing, Sanitized Input Summaries, CORS Guards
Release Date:	August 2026

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1. Executive Summary & DFD Methodology

ScamShield is an AI-powered cybersecurity platform designed to protect non-technical users from digital scams, financial fraud, phishing, KYC traps, and impersonation attacks across SMS, WhatsApp, Emails, Web URLs, Screenshots, and Audio Call Recordings.

This document models how information moves into, through, and out of ScamShield using standard **Gane & Sarson DFD Notation**:

Symbol Type	Gane & Sarson Shape	ScamShield System Mapping
External Entity	Double Square Box	Users (Web/PWA/Android), External OCR/AI APIs, Cybercrime Authorities (1930 portal)
Process	Rounded Rectangle / Circle	Numbered routines (1.0 Data Ingestion, 3.0 Risk Engine, etc.) executing business logic
Data Store	Open-ended Rectangle	DS1 (Scans DB), DS2 (Quiz/Stats DB), DS3 (Threat Rules), DS4 (Upload Storage)
Data Flow	Labeled Directional Arrow	Data payloads (Raw Text, OCR tokens, Risk Scores 0-100, Highlighted HTML)

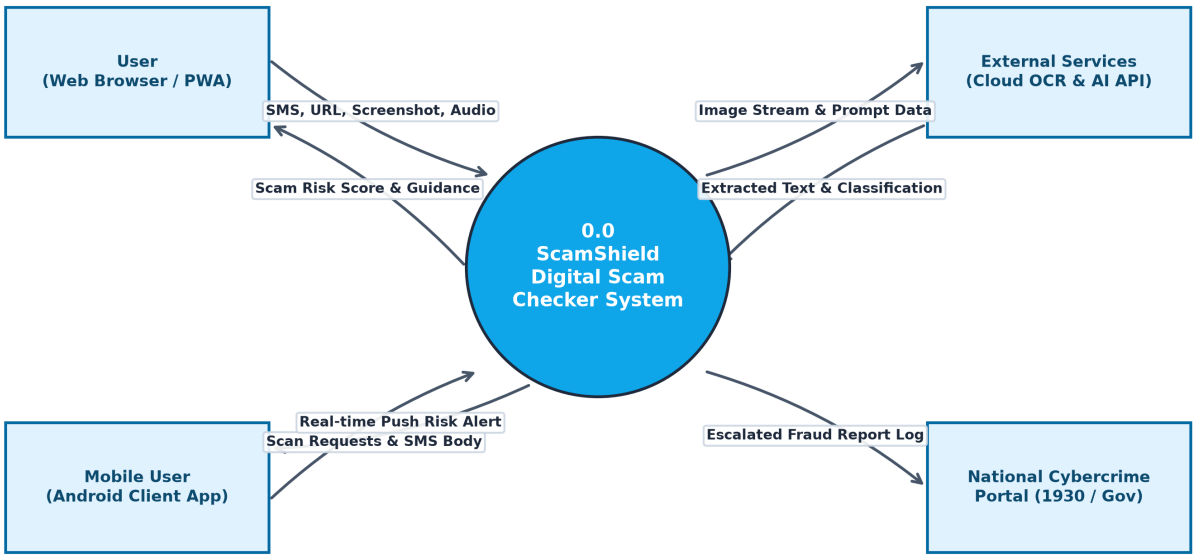
High-Level Data Transformation Flow:

- **Ingestion Phase:** User inputs raw content (Message text, URL string, Screenshot image, or Audio clip).
- **Feature Extraction:** Images undergo multi-tier OCR; audio undergoes speech transcription; text and URLs are tokenized.
- **Risk Calculation:** Pattern matching evaluates indicator weights against DS3 Threat Dictionary. Total weight is capped at 100.
- **Result Delivery:** Formatted risk level (LIKELY SAFE, SUSPICIOUS, HIGH RISK), warning signs, and suspicious phrase highlights are returned to the user.
- **Audit & Persistence:** Sanitized scan record is saved to DS1 (Scans DB) for user history and dashboard analytics.

2. Level 0: Context Diagram (Process 0.0)

The Level 0 Context Diagram establishes the highest-level operational boundary of the ScamShield system, identifying all external actors and primary data interactions.

DFD Level 0: Context Diagram (ScamShield Platform)



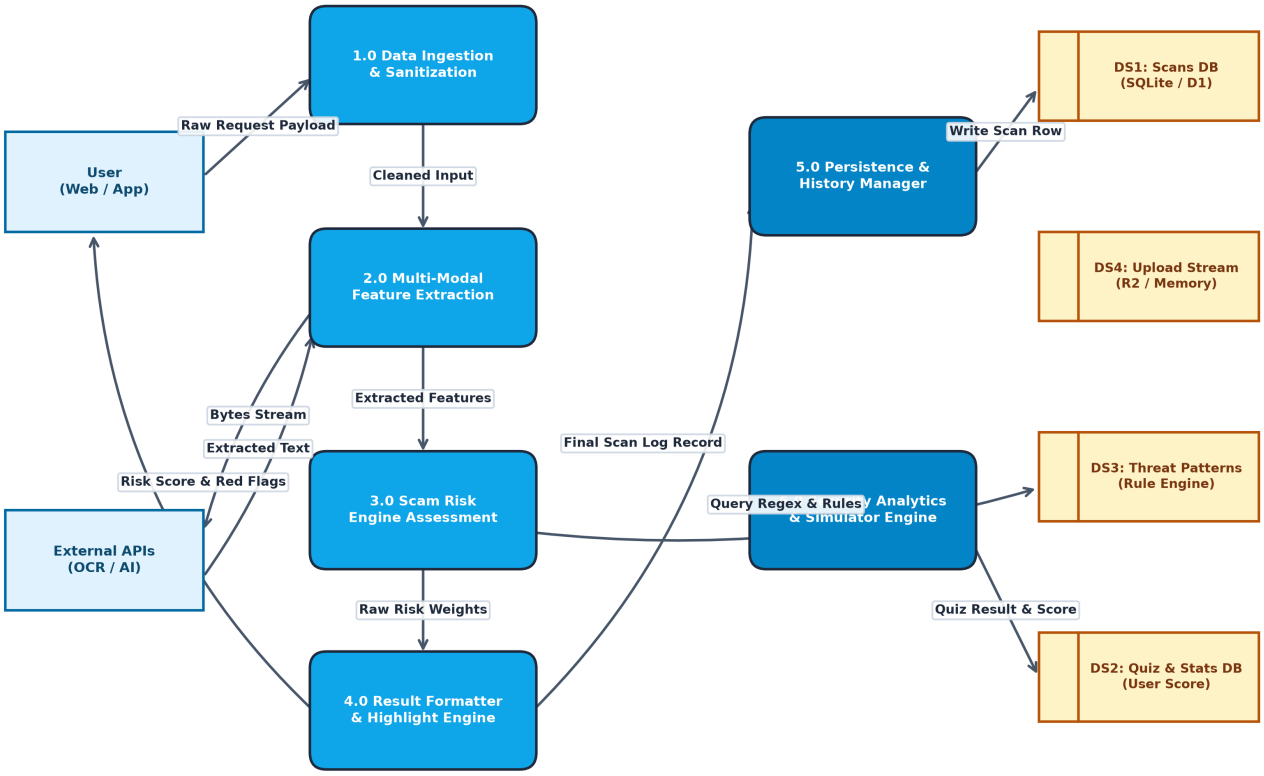
Level 0 Data Dictionary Table:

Flow Name	Source	Destination	Data Payload Description
Raw Input Payload	User (Web / PWA)	Process 0.0	SMS text, URL string, Screenshot image stream, Audio recording bytes
Scam Risk Report	Process 0.0	User (Web / PWA)	Risk score (0-100), risk level, warning signs, highlighted HTML text, guidance
Scan & Push Alert	Process 0.0	Mobile App User	Real-time danger rating & SMS scam category alert
OCR & AI Prompts	Process 0.0	External Services	Image byte stream, audio buffer, LLM classification prompt
Extracted Response	External Services	Process 0.0	Parsed OCR text, speech transcript, AI risk category
Fraud Escalation Log	Process 0.0	Cybercrime Portal	Escalation template with complaint metadata for official 1930 portal reporting

3. Level 1: System Level Data Flow Diagram

The Level 1 DFD decomposes Process 0.0 into 6 core functional processes, showing how data moves between ingestion, feature extraction, risk calculation, formatting, storage, and analytics.

DFD Level 1: System Level Data Flow Diagram



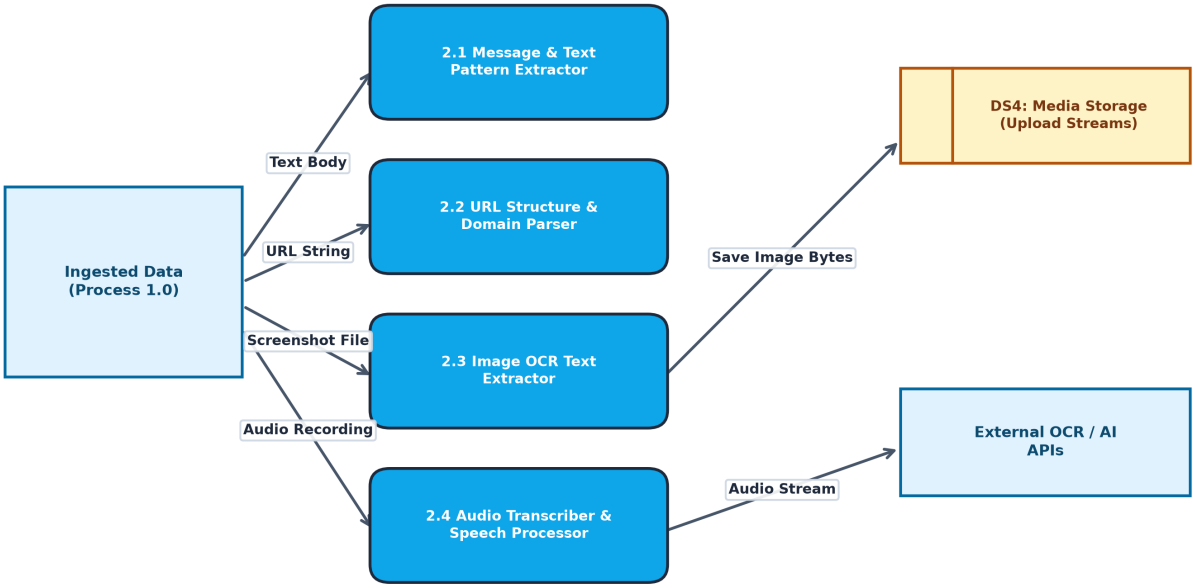
Level 1 Process Decomposition Directory:

Proc ID	Process Name	Input Flows	Output Flows
1.0	Data Ingestion & Sanitization	Raw Form / File Uploads	Sanitized Input Payload
2.0	Multi-Modal Feature Extraction	Sanitized Payload, OCR Bytes	Extracted Text & URL Features
3.0	Scam Risk Assessment Engine	Extracted Features, DS3 Rules	Indicator Score Weights & Category
4.0	Result Formatter & Highlight Engine	Raw Weights & Level	Formatted HTML Result, Final Log Record
5.0	Persistence & History Manager	Final Scan Log Record	DS1 SQL Insert / Query
6.0	Security Analytics & Simulator	Quiz Submission, Analytics Query	DS2 Score Updates, Dashboard Metrics

4. Level 2 Subsystem Decompositions (2.1 & 2.2)

Subsystem 2.1: Multi-Modal Feature Extraction (Decomposition of Process 2.0)

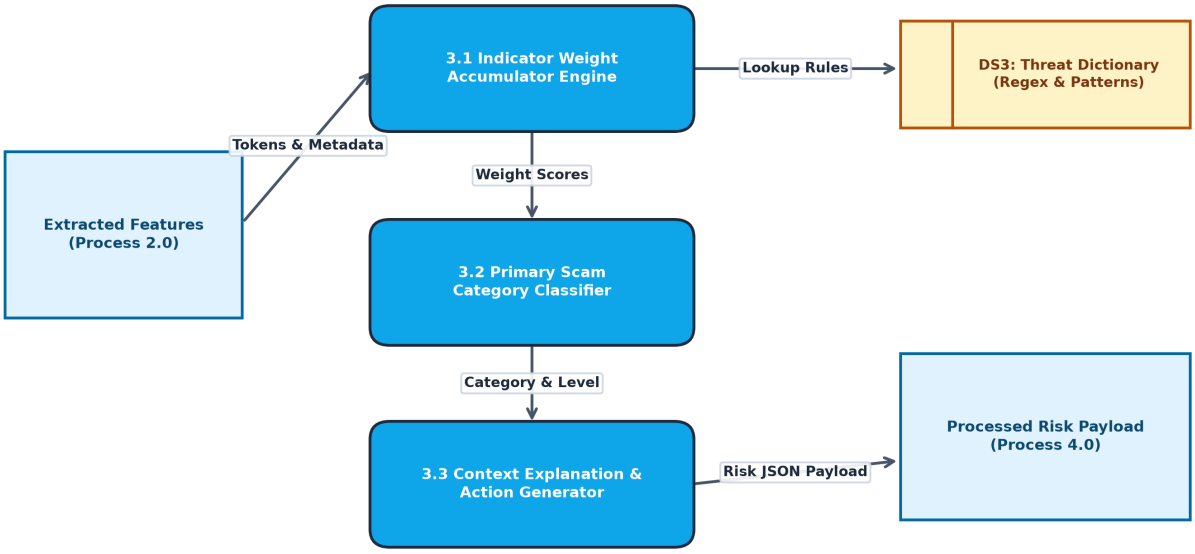
DFD Level 2.1: Multi-Modal Feature Extraction Subsystem (Process 2.0)



• **2.1 Text Extractor:** Tokenizes text messages and extracts suspicious regex keywords (e.g. 'blocked', 'urgent', 'winner').
• **2.2 URL Parser:** Resolves shortened URLs (bit.ly, tinyurl), checks domain age, IP address usage, and phishing keywords.
• **2.3 Screenshot OCR:** Converts uploaded screenshot images to text using in-memory Pytesseract / Cloud OCR.
• **2.4 Audio Transcriber:** Processes audio clips into text transcripts for scam keyword detection.

Subsystem 2.2: Risk Assessment & Classification (Decomposition of Process 3.0)

DFD Level 2.2: Risk Assessment & Classification Subsystem (Process 3.0)

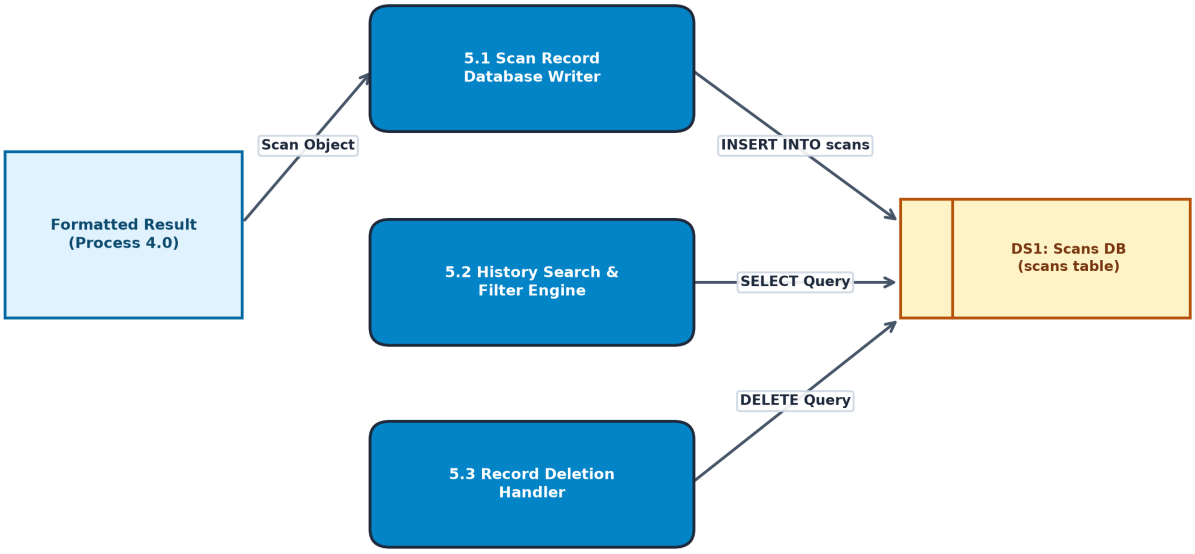


• **3.1 Indicator Weight Accumulator:** Queries DS3 Threat Dictionary and sums risk weights for all detected indicators (capped at 100).
• **3.2 Primary Category Classifier:** Determines primary scam type (e.g. KYC Scam, UPI Scam, Banking Scam, Phishing, Job Scam).
• **3.3 Explanation & Action Generator:** Builds human-understandable risk explanations and recommended safety actions based on risk tier.

5. Level 2 Subsystem Decompositions (2.3 & 2.4)

Subsystem 2.3: Data Persistence & History (Decomposition of Process 5.0)

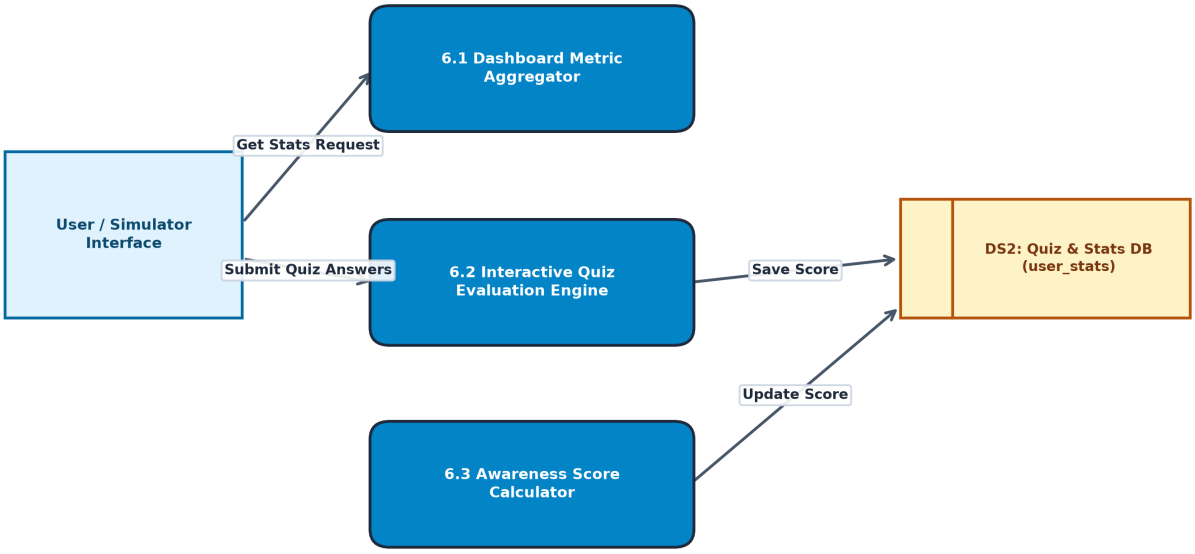
DFD Level 2.3: Data Persistence & History Subsystem (Process 5.0)



- **5.1 Scan Record Writer:** Serializes risk scores, category, warning signs JSON, and input summary into DS1 (`scans` table).
- **5.2 History Search & Filter:** Executes filtered SELECT queries on DS1 by search term, risk level, or date sorting.
- **5.3 Record Deletion Handler:** Performs secure deletion of individual scan records upon user request.

Subsystem 2.4: Security Analytics & Simulator (Decomposition of Process 6.0)

DFD Level 2.4: Analytics & Simulator Subsystem (Process 6.0)



- **6.1 Dashboard Metric Aggregator:** Calculates total scans, threat category distribution, and average risk score for `/dashboard`.
- **6.2 Interactive Quiz Evaluator:** Scores user responses in the 'Spot the Scam' simulator.
- **6.3 Awareness Score Calculator:** Updates overall cyber awareness index score in DS2 (`user_stats` table).

6. Data Store Specifications & SQL Schemas

ScamShield utilizes Cloudflare D1 serverless SQL database with local SQLite fallback. Below are the complete schema definitions for all system data stores.

DS1: Scans Database (`scans` Table Schema)

Field Name	Data Type	Constraints	Description / Content
`id`	INTEGER	PRIMARY KEY AUTOINCREMENT	Unique scan identifier
`created_at`	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Scan execution timestamp
`scan_type`	TEXT	NOT NULL	'message', 'url', 'screenshot', 'call'
`risk_score`	INTEGER	NOT NULL (0-100)	Calculated scam risk score
`risk_level`	TEXT	NOT NULL	'LIKELY SAFE', 'NEEDS VERIFICATION', 'SUSPICIOUS', 'HIGH RISK'
`scam_category`	TEXT	NOT NULL	'Banking Scam', 'UPI Scam', 'KYC Scam', 'Phishing', etc.
`warning_signs_json`	TEXT (JSON)	NULLABLE	JSON array of detected red flags
`explanation`	TEXT	NULLABLE	Contextual risk explanation paragraph
`input_summary`	TEXT	NULLABLE	Sanitized, truncated summary of analyzed content

DS2: Quiz & User Stats Database (`quiz_results` & `user_stats` Schemas)

Table	Field	Type	Description
`quiz_results`	`score`, `total_questions`	INTEGER	Number of correct answers and total questions in quiz run
`quiz_results`	`accuracy`	REAL	Accuracy percentage (0.0 to 100.0)
`user_stats`	`awareness_score`	INTEGER	Aggregated user cyber safety awareness score (default 85)
`user_stats`	`total_scans`	INTEGER	Total number of scans completed across all channels

7. Data Security, Privacy & Boundary Governance

To ensure compliance with cybersecurity best practices and protect sensitive user data, ScamShield enforces strict data boundary controls across all DFD processes:

1. Ephemeral In-Memory File Processing (Zero Disk Dependency)

Screenshots and audio recordings uploaded to Process 1.0/2.0 are converted into in-memory byte streams (`io.BytesIO`) during OCR and audio transcription. No persistent files are written to ephemeral disk storage in production.

2. Input Sanitization & Privacy Truncation

Before persisting scan logs to DS1 (`scans` table), input strings undergo HTML escaping and sanitization. Full message contents are truncated to safe summaries (`input_summary`), ensuring credentials, OTPs, or passwords accidentally submitted by users are not logged.

3. HTTP Security Headers & Cross-Origin Resource Sharing (CORS)

All API endpoints enforce mandatory security headers (`X-Content-Type-Options: nosniff`, `X-Frame-Options: SAMEORIGIN`, `X-XSS-Protection: 1; mode=block`). Requests from unauthorized origins are blocked.

4. Escalation Protocol & Helpline Integration

High-risk scans (Score >= 75) trigger direct emergency recommendations guiding users to immediately dial the **National Cyber Crime Helpline (1930)** or file complaints at **cybercrime.gov.in**.

Document Sign-off:
Architect: *ScamShield Engineering Team*
Status: *Approved & Verified*